Cybersecurity & Ethical Hacking / Bug Bounty Career Track

Course Syllabus — Module-Based Structure

Module No.	Title
Module 1	Cybersecurity Fundamentals
Module 2	Networking Fundamentals & Linux for Cybersecurity
Module 3	Reconnaissance & Footprinting
Module 4	Web Security Fundamentals
Module 5	Web Application Penetration Testing — OWASP Top 10
Module 6	Network Security & Exploitation
Module 7	Privilege Escalation & Post-Exploitation
Module 8	Cloud Security & Security Monitoring
Module 9	Bug Bounty & Executive Reporting
Module 10	Security Projects, Certification & Career Readiness

Course Overview

This Cybersecurity & Ethical Hacking course is designed for learners who want to build practical, job-ready offensive security skills — from first principles through to earning income via bug bounty programs. The course begins with foundational security and networking concepts and progressively builds toward reconnaissance, web application penetration testing, network exploitation, privilege escalation, cloud security, and professional-grade vulnerability reporting. Learners will gain exposure to the OWASP Top 10, industry-standard tooling, and the same methodology used by professional penetration testers and bug bounty hunters in real-world engagements.

The course blends theoretical grounding with hands-on practice. Learners will scan and enumerate live lab targets, exploit and remediate common vulnerabilities in a safe environment, escalate privileges on vulnerable machines, and write professional vulnerability reports. A capstone penetration test ties every concept together into a portfolio-ready deliverable.

Duration

The course consists of 10 modules with an estimated total learning duration of 62–74 hours, depending on instructor-led or self-paced delivery. Each module combines theory, hands-on labs, activities, and assessments.

Format

The course follows a mix of lectures, hands-on labs, capture-the-flag style exercises, guided walkthroughs, and a capstone project. Lectures introduce concepts; labs let learners attack intentionally vulnerable machines and applications in a safe, isolated environment; the capstone brings everything together into a documented penetration test report.

Prerequisites

Basic computer literacy is required. Comfort using a command line is a plus but not required — Linux and networking fundamentals are introduced progressively from first principles in Module 2. No prior security experience is needed.

Target Audience

This course suits IT professionals, developers, students, and career-changers who want to build offensive security skills. It is also useful for those preparing for roles such as Penetration Tester, SOC Analyst, Bug Bounty Hunter, Application Security Engineer, and Junior Red Teamer.

Course Objectives

1. Explain Core Cybersecurity Concepts and the Threat Landscape
Learners will understand the CIA triad, common threat actors, attack vectors, and risk concepts, and how these map onto the cyber kill chain and MITRE ATT&CK framework.

2. Apply Networking and Linux Fundamentals for Security
Learners will understand TCP/IP, DNS, HTTP/HTTPS, ports and protocols, and essential Linux command-line skills used for traffic analysis, enumeration, and exploitation.

3. Perform Professional Reconnaissance and Footprinting
Learners will use OSINT, DNS enumeration, subdomain discovery, and network scanning to build a complete attack-surface profile of a target before any exploitation begins.

4. Identify and Exploit Web and OWASP Top 10 Vulnerabilities
Learners will study and practically exploit Cross-Site Scripting (XSS), SQL Injection, CSRF, broken access control, SSRF, and other OWASP Top 10 risks in a lab setting.

5. Exploit Network Services and Escalate Privileges
Learners will use industry-standard tools to exploit vulnerable services, crack credentials, and escalate from low-privileged access to full administrative control on Windows and Linux targets.

6. Assess Cloud Environments and Security Monitoring
Learners will identify common cloud misconfigurations, understand the shared responsibility model, and interpret SIEM/logging data to recognize suspicious activity.

7. Operate as a Professional Bug Bounty Hunter
Learners will read program scope, apply a repeatable recon-to-report methodology, and write clear, well-evidenced vulnerability reports that get accepted and rewarded.

Module 1 — Cybersecurity Fundamentals

Estimated Duration: 5–6 hours

Goal: Build the mental model of what security is and how attackers and defenders think.

1. Definition and Goals of Cybersecurity — Protecting confidentiality, integrity, and availability (the CIA triad) of systems, applications, and data.

2. Threat Landscape Overview — Common threat actors — script kiddies, organized crime, insider threats, nation-states — and their typical motivations.

3. Hacker Types and Ethics — White/grey/black hat distinctions, and the legal and ethical boundaries of authorized security testing.

4. The Cyber Kill Chain & MITRE ATT&CK — A structured way to think about attacker phases, from reconnaissance through to objectives, and how frameworks map real-world techniques.

5. Core Terminology — Vulnerability, exploit, payload, zero-day, and how CVE/CVSS scoring communicates severity.

6. Lab Environment Setup — Installing VirtualBox/VMware, Kali Linux, and intentionally vulnerable target VMs to build a personal practice lab.

7. Legal & Ethical Considerations — Scope, written authorization, and responsible disclosure — the boundaries that separate ethical hacking from a crime.

Learning Outcomes: Learners can describe core security principles, common attack categories, and the legal boundaries of authorized testing.

Hands-On Lab: Build a personal pentesting lab environment with an attacker VM and at least one vulnerable target VM.

Module 2 — Networking Fundamentals & Linux for Cybersecurity

Estimated Duration: 6–7 hours

Goal: Master the networking and operating-system foundations every hacker needs.

1. TCP/IP & OSI Models — How data moves across networks, explained practically rather than theoretically.

2. IP Addressing, Subnetting & Routing — Reading and reasoning about network ranges to understand where a target sits.

3. TCP vs UDP & Common Ports — The three-way handshake and the ports/services attackers look for first.

4. DNS, HTTP/HTTPS & Packet Fundamentals — How domain resolution and web requests work at the protocol level.

5. Traffic Analysis with Wireshark — Capturing and reading live network traffic to spot cleartext credentials and anomalies.

6. Linux Essentials — Filesystem structure, permissions, users/groups, and the shell as your primary interface.

7. Bash Scripting for Automation — Writing small scripts to automate repetitive recon and exploitation tasks.

8. Command-Line Toolkit — Practical use of grep, awk, sed, netcat, and curl for security work.

Learning Outcomes: Learners can navigate Linux confidently, explain core networking/HTTP concepts, and read live traffic captures.

Hands-On Lab: Capture and analyze live traffic with Wireshark; complete a set of Linux permissions and privilege exercises.

Module 3 — Reconnaissance & Footprinting

Estimated Duration: 5–6 hours

Goal: Learn how attackers gather intelligence before ever touching an exploit.

1. Passive vs Active Reconnaissance — The trade-off between staying invisible and gathering deeper information.

2. OSINT Techniques — Google dorking, WHOIS lookups, Shodan, and theHarvester for gathering public intelligence.

3. Subdomain Enumeration — Using Sublist3r, Amass, and certificate transparency logs (crt.sh) to map an organization's footprint.

4. DNS Enumeration — Zone transfers and record analysis to reveal hidden infrastructure.

5. Network Scanning with Nmap — Host discovery, port scanning, service/version detection, and script scanning.

6. Banner Grabbing & Fingerprinting — Identifying exact software versions to target known vulnerabilities.

7. Building an Attack Surface Map — Consolidating recon data into a structured profile that drives the rest of the engagement.

Learning Outcomes: Learners can independently run a full reconnaissance workflow and produce a structured attack-surface profile.

Hands-On Lab: Perform a complete recon workflow against a lab target and deliver a written attack-surface report.

Module 4 — Web Security Fundamentals

Estimated Duration: 6–7 hours

Goal: Understand how web applications work so you know exactly where they can break.

1. HTTP/HTTPS Deep Dive — Methods, headers, status codes, and cookies from a security-testing perspective.

2. Web Architecture — Client-server model, APIs, databases, and sessions, and where trust boundaries exist.

3. Browser Developer Tools — Inspecting requests, storage, and client-side code directly in the browser.

4. Burp Suite Fundamentals — Intercepting, modifying, and replaying requests using Proxy, Repeater, and Intruder.

5. Authentication vs Authorization — The critical distinction that underlies most access-control vulnerabilities.

6. Session Management & Cookie Security — Secure, HttpOnly, and SameSite attributes, and what happens when they're missing.

7. Same-Origin Policy & CORS — How browsers isolate content by origin and how misconfigured CORS breaks that isolation.

Learning Outcomes: Learners can intercept, read, and manipulate live web traffic and explain how session and origin security works.

Hands-On Lab: Intercept and manipulate live requests against a lab web application using Burp Suite.

Module 5 — Web Application Penetration Testing — OWASP Top 10

Estimated Duration: 10–12 hours

Goal: Hands-on exploitation of the most critical and common web application vulnerabilities.

1. Broken Access Control — Exploiting IDOR and privilege-escalation flaws to access data and functions you shouldn't.

2. Cryptographic Failures — Identifying weak encryption and unnecessary exposure of sensitive data.

3. Injection (SQL & Command) — Manual SQL injection technique plus automated exploitation with sqlmap; OS command injection.

4. Insecure Design & Security Misconfiguration — Recognizing architectural flaws and default/verbose configurations that widen the attack surface.

5. Vulnerable & Outdated Components — Identifying and exploiting known CVEs in outdated libraries and frameworks.

6. Authentication Failures — Brute-force and credential-stuffing attacks against weak login flows.

7. Server-Side Request Forgery (SSRF) — Tricking a server into making unintended requests on the attacker's behalf.

8. Cross-Site Scripting (XSS) — Reflected, stored, and DOM-based XSS, and how each is triggered and evidenced.

9. Cross-Site Request Forgery (CSRF) — Forcing authenticated users to unknowingly submit malicious requests.

10. File Upload & Path Traversal — Bypassing upload filters and traversing directories to reach unauthorized files.

Learning Outcomes: Learners can independently identify, exploit, and explain the impact of every OWASP Top 10 vulnerability class.

Hands-On Lab: Complete guided exercises against DVWA, OWASP Juice Shop, and PortSwigger Web Security Academy labs covering every vulnerability class above.

Module 6 — Network Security & Exploitation

Estimated Duration: 7–8 hours

Goal: Move from web applications to full network-level attacks.

1. Deep Network Enumeration — Advanced Nmap usage and service-specific enumeration techniques.

2. Vulnerability Scanning — Running and interpreting results from Nessus/OpenVAS-style scanners.

3. Metasploit Fundamentals — Searching, configuring, and firing exploit modules against vulnerable services.

4. Exploiting Common Services — Attacking misconfigured SMB, FTP, SSH, and RDP services.

5. Password Attacks — Brute-force and dictionary attacks using Hydra, John the Ripper, and Hashcat.

6. Man-in-the-Middle Attacks — ARP spoofing and intercepting traffic on a local network.

7. Wireless Security Basics — An overview of WPA2 handshake capture and cracking concepts.

8. Firewall & IDS/IPS Evasion Basics — Understanding how defenses detect attacks and basic techniques to reduce noise.

Learning Outcomes: Learners can chain scanning, exploitation, and credential attacks to gain initial access on vulnerable network targets.

Hands-On Lab: Fully exploit vulnerable lab machines end-to-end, from scan to shell (HackTheBox/TryHackMe-style targets).

Module 7 — Privilege Escalation & Post-Exploitation

Estimated Duration: 6–7 hours

Goal: What happens after initial access — go deeper, escalate, and understand persistence.

1. Windows Privilege Escalation — Exploiting misconfigured services, tokens, and registry settings to reach Administrator.

2. Linux Privilege Escalation — Abusing SUID binaries, cron jobs, and kernel exploits to reach root.

3. Automated Enumeration Scripts — Using LinPEAS and WinPEAS to rapidly surface escalation paths.

4. Pivoting & Lateral Movement — Using a compromised host as a foothold to reach otherwise unreachable systems.

5. Persistence Mechanisms — How attackers maintain access after the initial compromise (for detection awareness).

6. Introduction to Active Directory Attacks — Kerberoasting and pass-the-hash concepts at an introductory level.

Learning Outcomes: Learners can take low-privileged access on a vulnerable machine and reliably escalate to full administrative control.

Hands-On Lab: Complete a full privilege-escalation chain on a vulnerable VM, from low-privilege shell to root/admin.

Module 8 — Cloud Security & Security Monitoring

Estimated Duration: 6–7 hours

Goal: Extend offensive skills into modern cloud environments and understand the defender's view.

1. Cloud Shared Responsibility Model — What AWS/Azure/GCP secure versus what the customer is responsible for.

2. Common Cloud Misconfigurations — Open S3 buckets, overly permissive IAM policies, and exposed metadata endpoints.

3. Cloud Recon Tools — Using ScoutSuite and Prowler-style tools to audit a cloud account's configuration.

4. Container & Kubernetes Security Basics — Recognizing common container escape and misconfiguration risks.

5. SIEM Concepts — Log aggregation, correlation, and alerting from a defender's perspective.

6. Monitoring Tools Overview — An introduction to the ELK stack and Splunk-style log analysis.

7. Incident Response Lifecycle — Preparation, detection, containment, eradication, and recovery, at a high level.

Learning Outcomes: Learners can identify common cloud misconfigurations and interpret basic SIEM/monitoring output.

Hands-On Lab: Identify a deliberately misconfigured cloud storage bucket and review a set of sample SIEM alerts for signs of compromise.

Module 9 — Bug Bounty & Executive Reporting

Estimated Duration: 5–6 hours

Goal: Turn technical findings into real-world impact — and income.

1. How Bug Bounty Programs Work — The mechanics of HackerOne, Bugcrowd, and Synack-style platforms.

2. Choosing Programs & Reading Scope — Selecting targets that match your skill level and staying strictly within authorized scope.

3. Recon-to-Report Workflow — A repeatable methodology tailored specifically for bug bounty hunting efficiency.

4. Writing Effective Vulnerability Reports — Structuring a report with a technical section and a non-technical executive summary.

5. CVSS Scoring & Severity Justification — Defending your assessed severity with evidence, not just a gut feeling.

6. Proof-of-Concept Best Practices — Creating clear, minimal, reproducible PoCs that triagers can validate quickly.

7. Responsible Disclosure & Triage Communication — Professional communication that keeps reports moving instead of getting closed as N/A.

8. Avoiding Duplicates & Common Rejections — Practical habits that reduce wasted effort and improve acceptance rate.

Learning Outcomes: Learners can independently take a finding from discovery to a submitted, well-evidenced, professional report.

Hands-On Lab: Write a full professional vulnerability report — technical detail plus executive summary — from a lab finding.

Module 10 — Security Projects, Certification & Career Readiness

Estimated Duration: 6–8 hours

Goal: Consolidate everything into a portfolio and prepare for the job market.

1. Capstone Project — A full penetration test of a lab environment, from recon through to a final written report.

2. Building a Portfolio — Publishing CTF write-ups, lab walkthroughs, and tools on GitHub to show real capability.

3. Certification Landscape — An overview of CompTIA Security+, eJPT, PNPT, OSCP, and CEH, and how they differ.

4. Choosing a Certification Path — Matching a certification roadmap to your specific career goal.

5. Resume & LinkedIn Optimization — Positioning lab work and CTF experience effectively for security roles.

6. Interview Preparation — Practicing scenario-based technical questions alongside behavioral interview prep.

7. Career Path Exploration — Pentester, SOC Analyst, AppSec Engineer, and freelance Bug Bounty Hunter as distinct paths.

8. Continued Learning — Staying current with new CVEs, communities, and conferences after the course ends.

Learning Outcomes: Learners leave with a completed capstone penetration-test report, a public portfolio, and a security-focused resume.

Hands-On Lab: Deliver a complete, portfolio-ready penetration test report as the course's final capstone deliverable.

Assessment Methods

- Module quizzes (knowledge checks)
- Hands-on labs (graded on completion and correctness)
- Vulnerability report write-ups
- Final capstone penetration test and report

Tools & Platforms Used

Kali Linux, Nmap, Burp Suite, Wireshark, Metasploit, sqlmap, Hydra, John the Ripper, Hashcat, LinPEAS/WinPEAS, ScoutSuite, DVWA, OWASP Juice Shop, PortSwigger Web Security Academy.

Suggested Practice Platforms: TryHackMe, HackTheBox, PortSwigger Web Security Academy, VulnHub.

Certification Pathways Aligned With This Course

- CompTIA Security+ — foundational, vendor-neutral entry point
- eLearnSecurity Junior Penetration Tester (eJPT) — practical entry-level pentest cert
- Practical Network Penetration Tester (PNPT) — scenario-based, hands-on assessment
- Offensive Security Certified Professional (OSCP) — advanced, industry-respected follow-on path
- Certified Ethical Hacker (CEH) — broad, widely-recognized foundational credential

Key Technologies & Platforms Reference

Category	Technologies
Operating System	Kali Linux, Bash scripting
Networking	TCP/IP, DNS, HTTP/HTTPS, Wireshark
Reconnaissance	Nmap, Amass, Sublist3r, Shodan, theHarvester
Web Security Testing	Burp Suite, sqlmap, DVWA, OWASP Juice Shop
Exploitation	Metasploit, Hydra, John the Ripper, Hashcat
Post-Exploitation	LinPEAS, WinPEAS
Cloud Security	AWS, Microsoft Azure, Google Cloud Platform, ScoutSuite
Monitoring	SIEM concepts, ELK stack / Splunk-style log analysis
Bug Bounty Platforms	HackerOne, Bugcrowd, Synack

Total: 10 Modules | ~62–74 hours | Culminates in a portfolio-ready penetration test report